

WASP-43b

R-0391

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250306 · created 2026-07-19T00:09:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460740.8013 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.057
Transit depth (Rp/Rs)^2	2.6 +/- 1.83 [%]
Orbital Inclination (inc)	78.93 +/- 2.94
Airmass coefficient 1 (a1)	55.64 +/- 32.4
Airmass coefficient 2 (a2)	-3.0 +/- 0.93
Scatter in the residuals of the lightcurve fit is	52.63 %
Best Comparison Star	#1 - [229, 412]
Optimal Aperture	3.27
Optimal Annulus	5.27
Transit Duration (day)	0.032 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.161 $\pm$ 0.057 vs 0.1594 (deviation 0.02 σ)
Duration fit vs published	0.032 d vs 0.0512 d (deviation 0.69 σ)
Mid-time O–C	6.1 min
Depth SNR	1.9
Residual scatter	52.63 %

Light curve

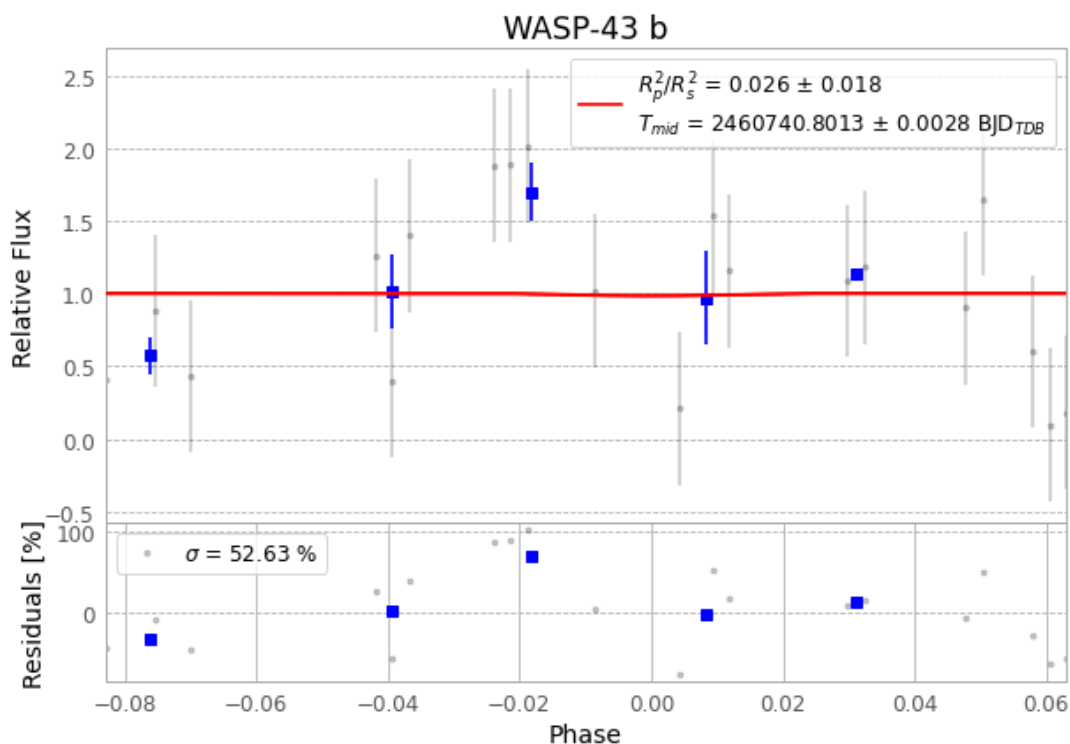


Figure 1 — FinalLightCurve_WASP-43 b_2025-03-05.png (SHA-256 e1c56ea8d6dca8cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [394, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 19b2e8fa500da2b3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

64 FITS frames (42 MB), WASP-43250306052718.FITS → WASP-43250306083615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250306.log (e79ee25a8e2f...), FinalLightCurve_WASP-43 b_2025-03-05.png (e1c56ea8d6dc...), FinalLightCurve_WASP-43 b_2025-03-05.csv (28d7f09ff830...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 00:09Z.